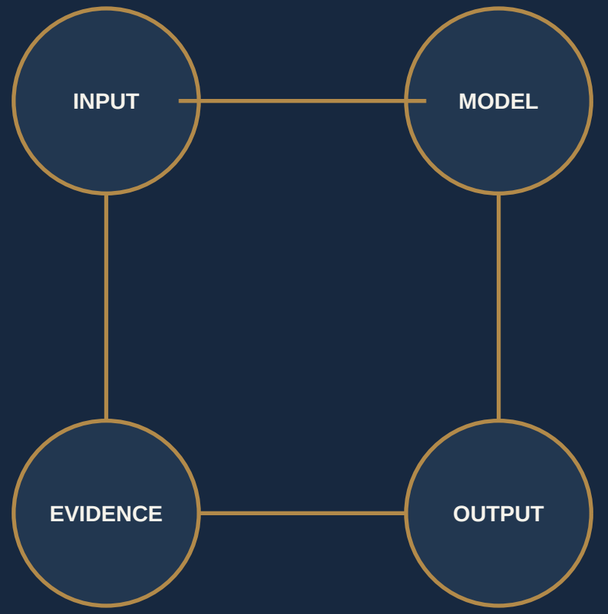
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Path Planning

AI.408 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define configuration spaces.
- Apply key methods including graph search.
- Build or analyze a practical system involving dijkstra.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Configuration spaces

02

METHODS

Graph search

03

SYSTEMS

Dijkstra

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Configuration spaces

Configuration spaces

02 · Graph search

Graph search

03 · Dijkstra

Dijkstra

04 · Sampling-based planning

Sampling-based planning

05 · Trajectory generation and coll

Trajectory generation and collision avoidance

06 · Mandatory practical: Implement

Mandatory practical: Implement several path-planning algorithms and compare their performance

Core ideas, made explicit

AI.408 • Distinguish definitions, assumptions and evidence

- Configuration spaces
- Graph search
- Dijkstra
- Sampling-based planning
- Trajectory generation and collision avoidance

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.408 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Implement several path-planning algorithms and compare their performance

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving configuration spaces.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using configuration spaces.
- Implement or simulate a baseline using graph search.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.